# Experimental Validation of a Corrugated Membrane for Large-Scale Liquid Hydrogen Cargo Containment

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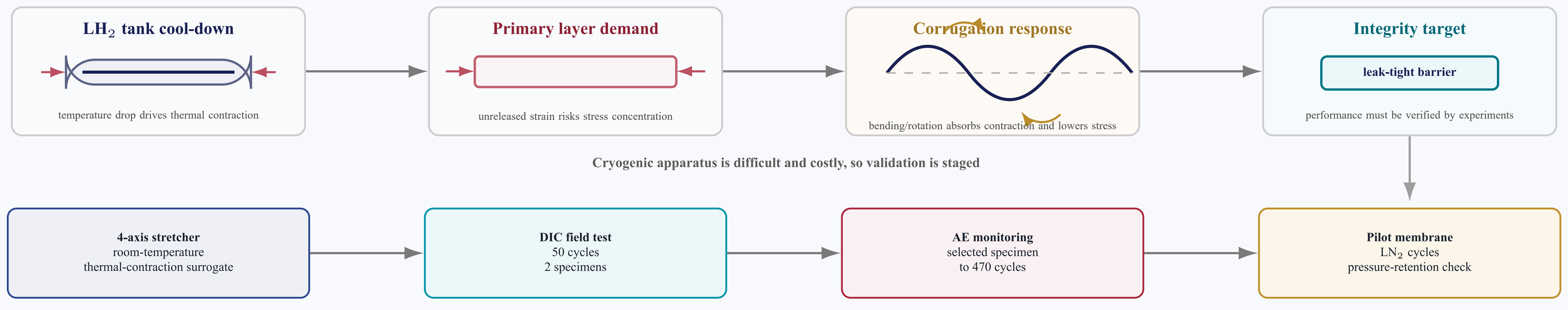
ICEC30 / ICMC2026
June 22–26, 2026
Daejeon Convention Center (DCC), Daejeon, Korea

## Introduction

Large-scale liquid hydrogen ($LH_2$) transport requires thin metallic containment barriers that remain gas-tight while accommodating thermal contraction and restrained in-plane deformation. Corrugated 316L stainless-steel membranes offer a promising solution because their geometry can redistribute imposed strain and reduce local deformation demand in nominally planar regions. Although the target application is cryogenic, room-temperature mechanical testing is used here to isolate the geometry-driven deformation-accommodation mechanism under controlled displacement loading, rather than to reproduce the full $LH_2$ service environment. Cryogenic integrity is examined separately through liquid-nitrogen ($LN_2$) cycling of a 0.2 $m^3$ integrated demonstrator, while full mechanical qualification at $LH_2$ temperature remains a subsequent validation step.

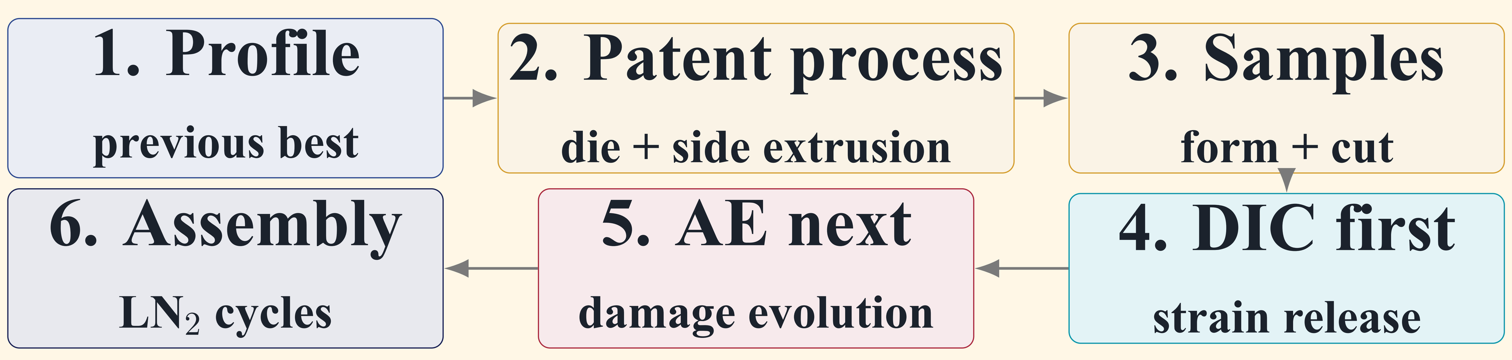
This study experimentally evaluates the corrugated membrane concept using full-field digital image correlation (DIC) under tensile loading, acoustic-emission-monitored cyclic testing, and $LN_2$ thermal cycling with pressure-retention and leak-check assessment. The results clarify how corrugation features accommodate deformation and provide preliminary evidence for assembly-level integrity, supporting further development of corrugated membrane systems for large-scale $LH_2$ cargo containment.

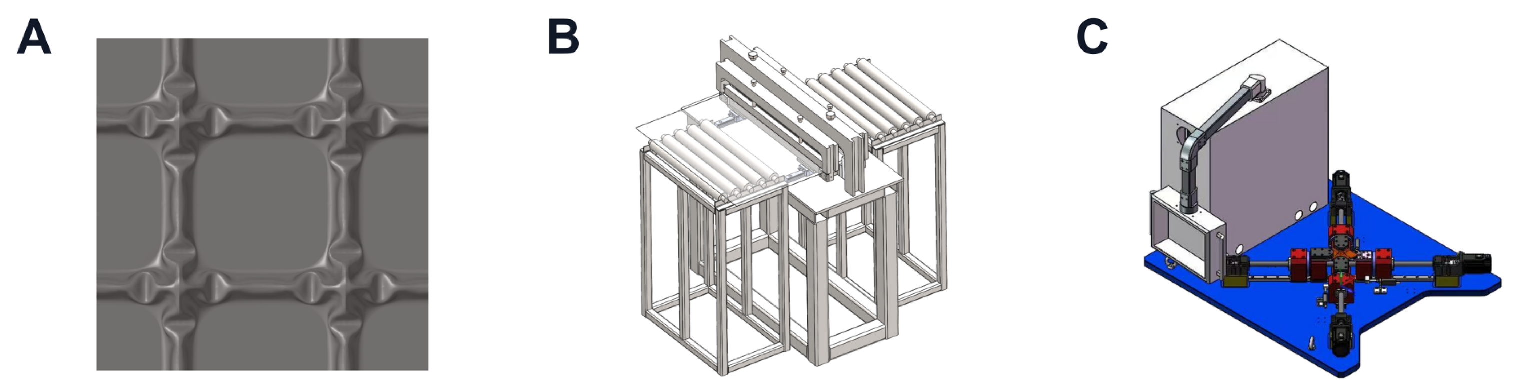



Conceptual schematic: cool-down contraction creates the primary-layer displacement demand; corrugations absorb the demand by geometric compliance; room-temperature DIC and AE tests screen strain redistribution and damage evolution before $LN_2$ cycling plus pressure-retention checks verify pilot-membrane integrity.

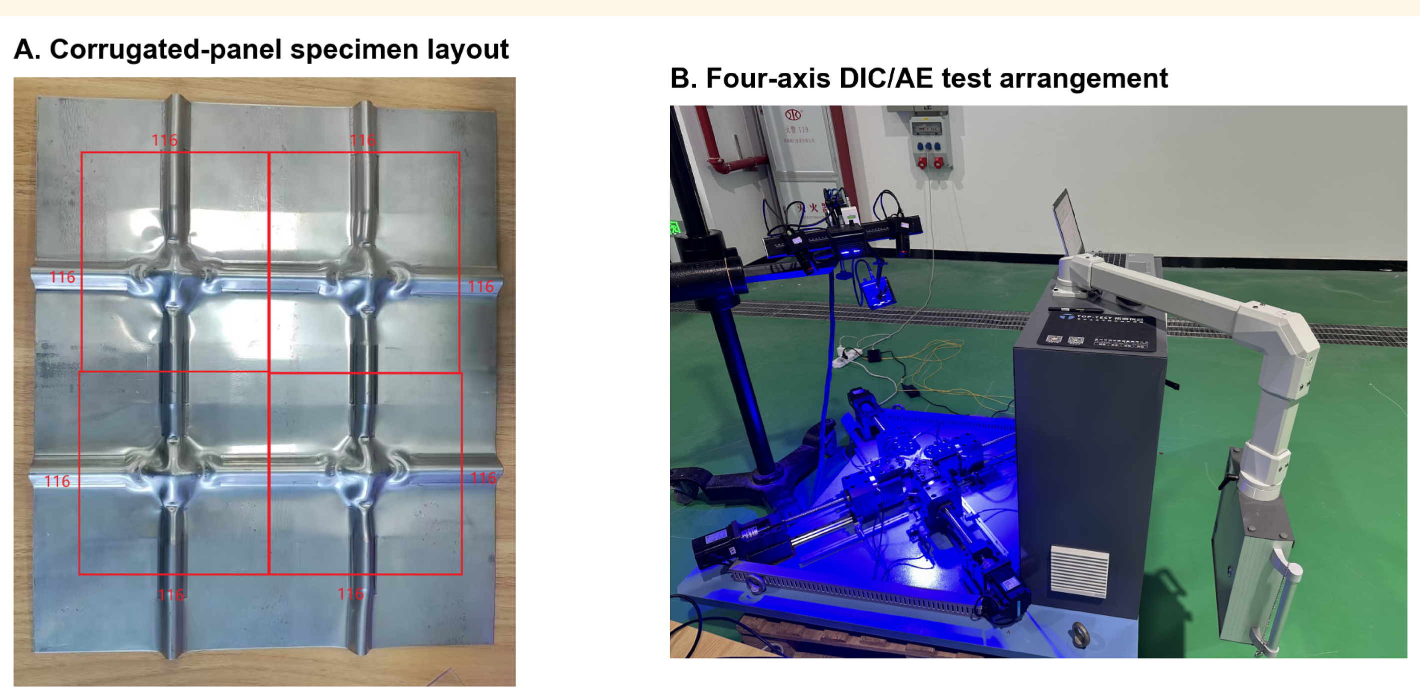
## M. Methods: Design-to-Test Logic Chain

| Starting point | Value | Why it matters |
|---|---|---|
| **Selected profile** | $e_r = 1.0, e_w = 5.0, \lambda = 0.651$ | best-performing geometry from the previous design screen |
| **Sheet and material** | 316L, $t = 1.2$ mm | thin metallic primary-barrier candidate |
| **Patent basis** | CN118237450A | coupled downward pressing and lateral squeezing for equal-section cross corrugation |




Selected profile → patented processing machine → four-axis specimen platform. The process figure is used as design and preparation context, not as a dimensional CAD claim.



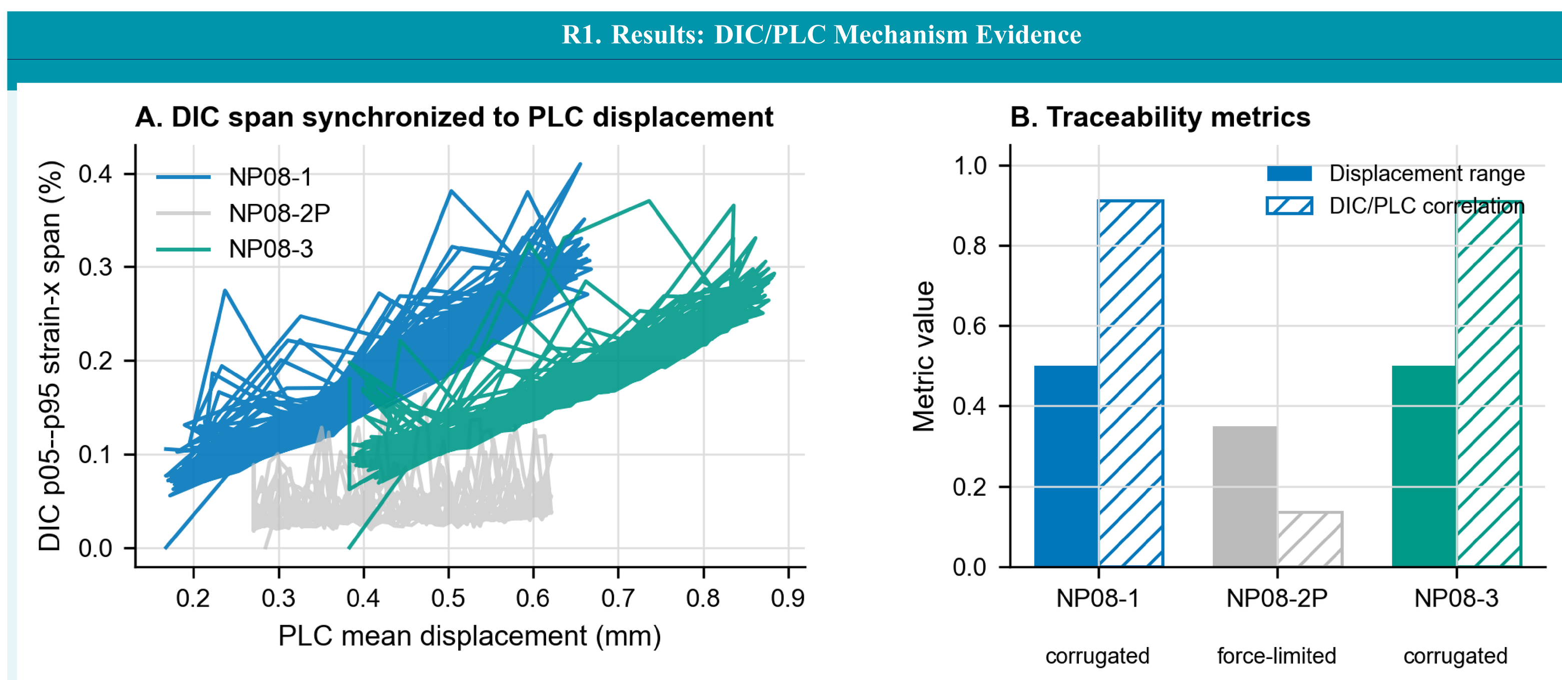

Specimen extraction, DIC field of view, and mechanical setup connect the formed profile to the DIC/AE tests.

| Experiment | Direct question | Measured output |
|---|---|---|
| **DIC/PLC** | How does corrugation release imposed strain? | synchronized strain span, geometry-fixed ROI hierarchy |
| **AE** | When does cyclic activity concentrate? | relabeled cycle windows and damage-evolution screen |
| **$LN_2$ assembly** | Does pressure-retention testing verify assembly integrity after thermal cycling? | pressure-hold and helium leak checks across five $LN_2$ cycles |

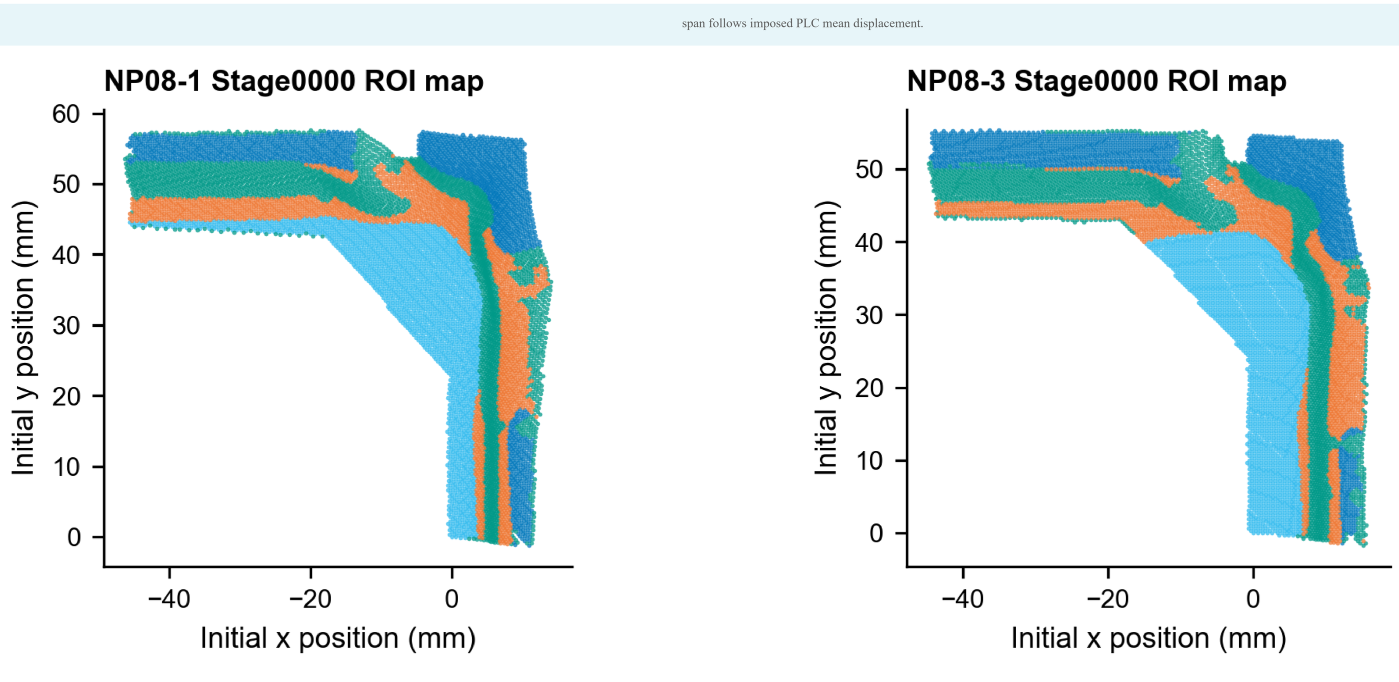
$$\bar{u}_k = \frac{1}{4}\sum_{a=1}^{4} u_{a,k}, \qquad S_i = \frac{dP_{95}(|\varepsilon_x|)_i}{d\bar{u}}, \qquad \Psi_i = A_i q_i^2$$

The evidence chain is deliberately ordered: profile and processing first, DIC mechanism second, AE damage-evolution screening third, and pressure-retention testing for assembly integrity last.

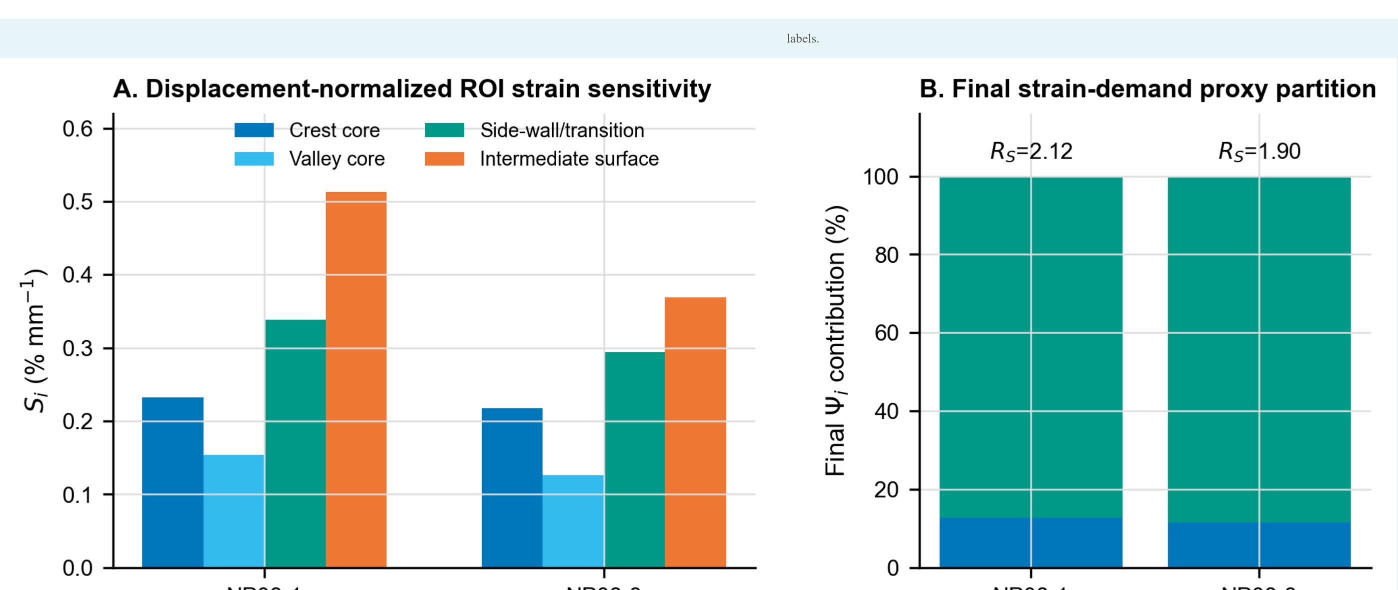
## R1. Results: DIC/PLC Mechanism Evidence




DIC strain span follows imposed PLC mean displacement.




Initial geometry defines persistent ROI labels.

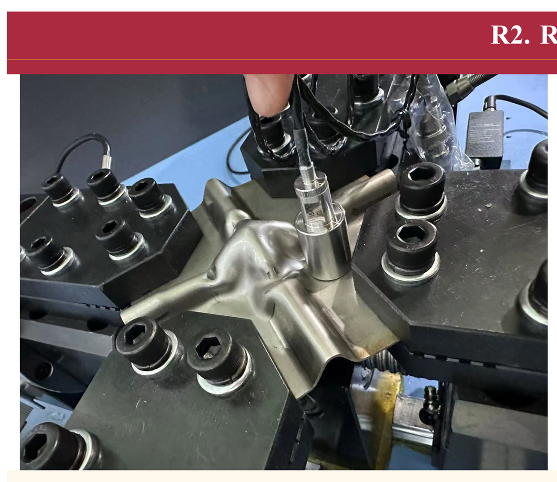



High-deformation ROIs carry most final strain demand.

| Metric | NP08-1 | NP08-3 | Interpretation |
|---|---|---|---|
| **Sensitivity ratio $R_S$** | 2.117 | 1.896 | high-ROI response is about 2x core response |
| **High-deformation area** | 54.28% | 53.96% | deformation path occupies about half the map |
| **Final strain-demand share** | 87.2% | 88.4% | most demand stays in compliant regions |

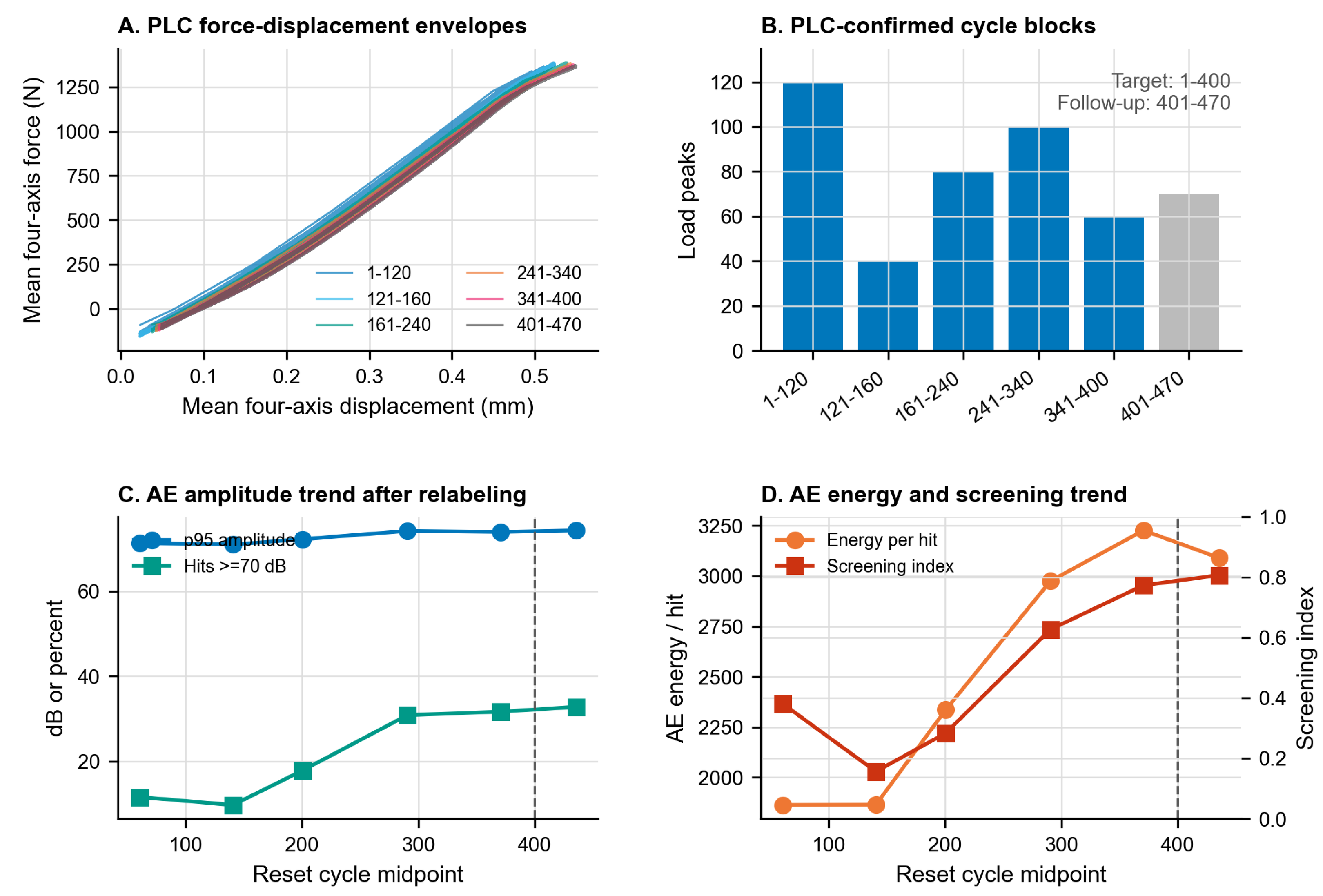
**Mechanism result.** Corrugated specimens show repeated ROI hierarchy: side-wall/transition and intermediate surfaces absorb the imposed displacement while crest/valley cores remain lower-demand regions.

## R2. Results: AE Activity and Assembly Integrity Checks

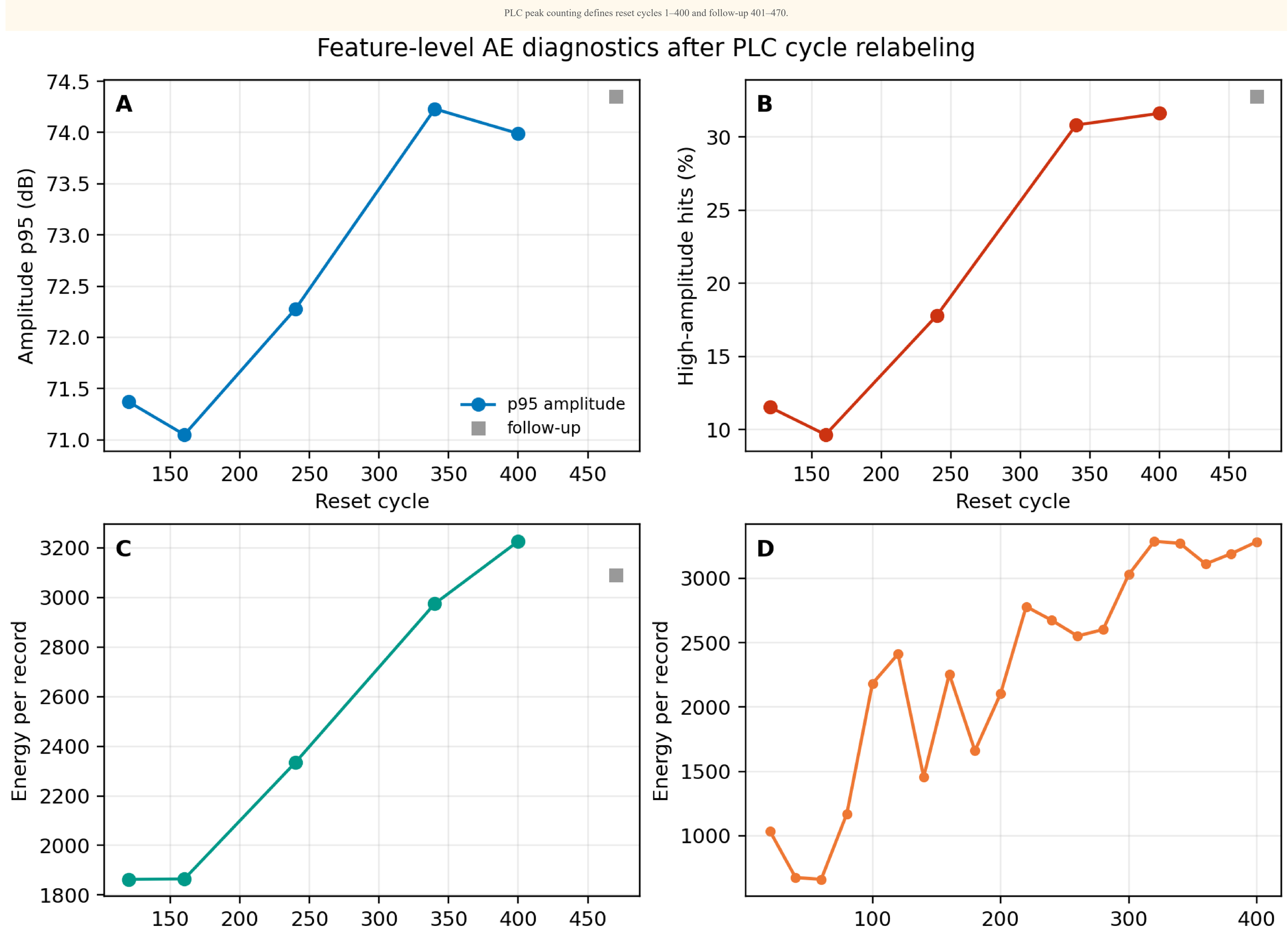


AE sensors were mounted during the same four-axis loading context.

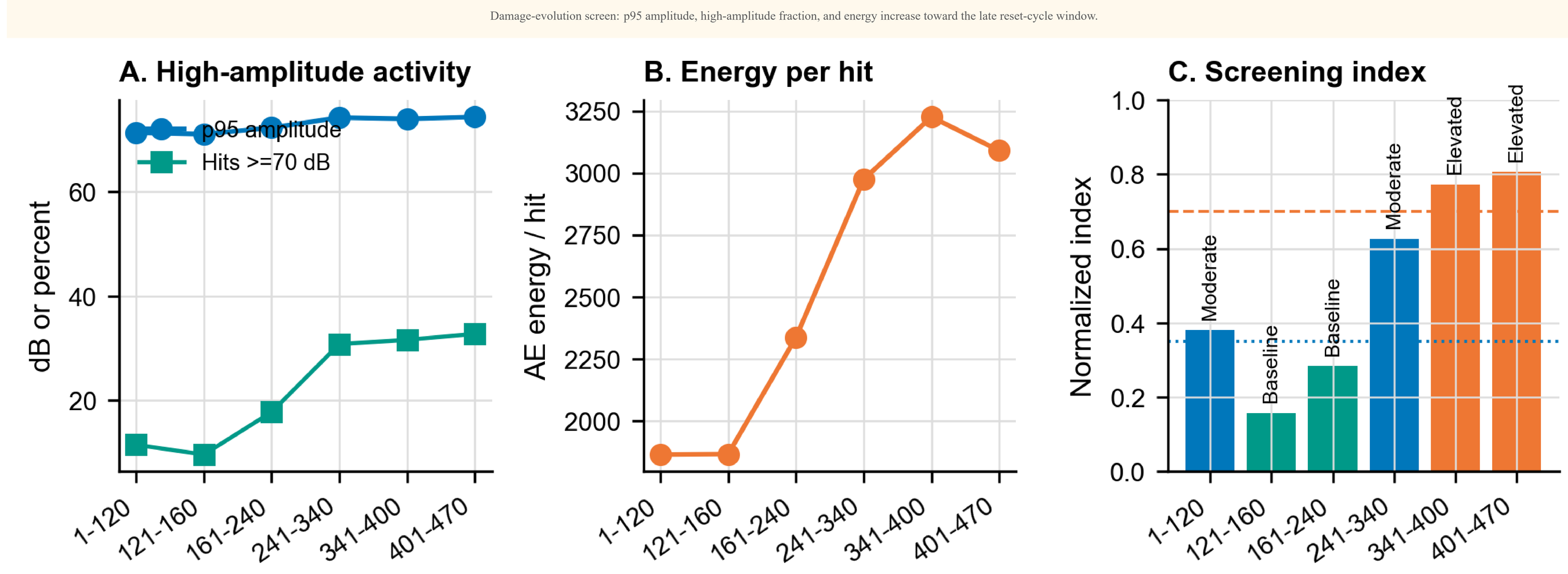
| Rule | Interpretive use |
|---|---|
| **Relabel cycles** | PLC peak counting resets the AE coordinate before interpretation. |
| **Track features** | amplitude, high-amplitude fraction, and energy indicate cyclic activity. |
| **Keep boundary** | feature trends screen damage evolution; they do not alone prove cracks. |




PLC peak counting defines reset cycles 1–400 and follow-up 401–470.




Damage-evolution screen: p95 amplitude, high-amplitude fraction, and energy increase toward the late reset-cycle window.

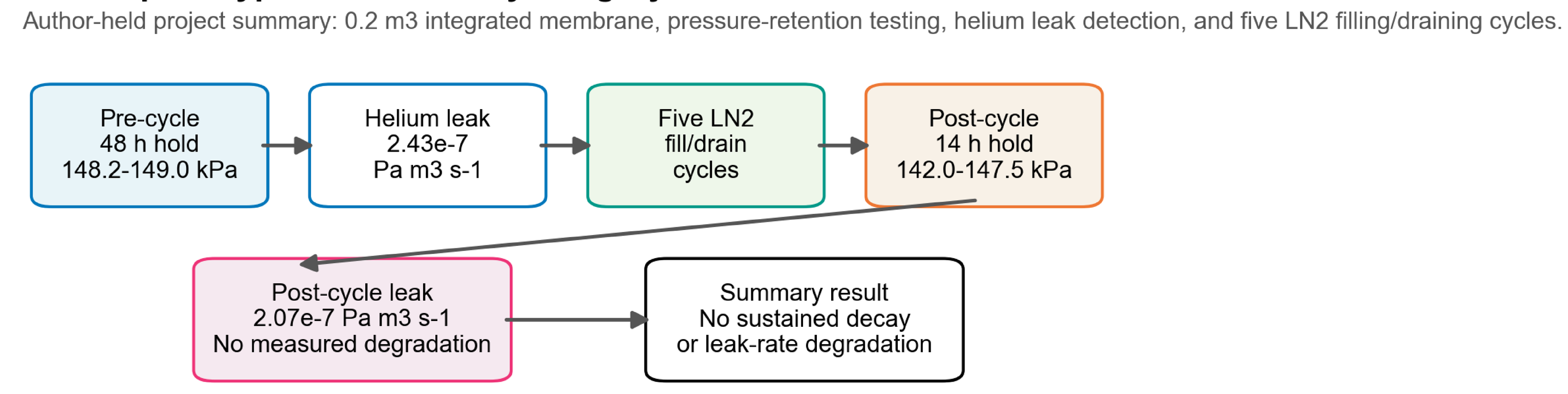



Screening index ranks cycles 341–400 as the inspection-priority window.

| 31.63% | $3.23 \times 10^3$ | 125 kHz | 1.17M |
|---|---|---|---|
| records above 70 dB | energy per hit | median frequency | feature records |

### Scaled-prototype LN2 assembly-integrity check

Author-held project summary: 0.2 m3 integrated membrane, pressure-retention testing, helium leak detection, and five LN2 filling/draining cycles.




Claim boundary: pressure-retention is the test method; the supported objective is preliminary assembly integrity, not LH2 qualification.

Pressure-hold and helium leak-rate checks are the test method; the objective is to verify assembled-membrane integrity before and after five $LN_2$ filling/draining cycles.

| Check | Result |
|---|---|
| **Pre-cycle hold** | 48 h at 148.2–149.0 kPa abs. |
| **Post-cycle hold** | 14 h at 142.0–147.5 kPa abs. |
| **Pre-cycle leak** | $2.43 \times 10^{-7}$ Pa $m^3$ $s^{-1}$ |
| **Post-cycle leak** | $2.07 \times 10^{-7}$ Pa $m^3$ $s^{-1}$ |

## D. Discussion: What the Evidence Supports

| Layer | Supports | Does not claim | Evidence panel |
|---|---|---|---|
| **DIC/PLC** | corrugation-driven strain redistribution | service lifetime | R1 |
| **AE** | late-cycle inspection-priority window | crack growth proof | R2-AE |
| **$LN_2$** | preliminary assembly integrity | $LH_2$ qualification | R2-$LN_2$ |

- The strongest evidence is mechanism evidence from synchronized DIC/PLC and ROI mapping.
- AE is useful after relabeling because it ranks when inspection attention should focus.
- $LN_2$ pressure-retention testing is necessary but not sufficient for final $LH_2$ service readiness.

## C. Conclusions and Next Gates

### Conclusions

1. Corrugation redistributes strain into traceable geometry-defined ROIs.
2. The late AE reset window gives an inspection-priority signal.
3. Pre/post $LN_2$ pressure and leak checks support preliminary assembly integrity.

### Next gates

1. Repeat coupon tests with ROI uncertainty.
2. Add cryogenic DIC and waveform-level AE.
3. Report full acceptance pressure/leak records under $LH_2$-relevant criteria.

| | |
|---|---|
| **ICEC/ICMC record** | P1-061, Poster Session I, Exhibition Hall (111+112), June 23, 2026, 14:00–16:00 |
| **Claim boundary** | The evidence supports staged validation under the tested conditions; completed $LH_2$ service qualification remains outside the present scope. |